

# **The Paradox of Digital Literacy: The Role of the Cyber Hygiene Gap in Increasing Social Engineering Susceptibility Among Young Adults in the UAE**

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***Abstract***—In the digital landscape of the United Arab Emirates, the assumption that "Digital Natives" are protected by technical fluency remains a critical vulnerability. This study investigates the Paradox of Digital Literacy, demonstrating a critical Knowledge-Behaviour Gap, where high technical proficiency fosters an overconfidence that invites, rather than prevents, cyber-victimisation. Utilising a multi-phase quantitative design, the research surveyed 113 respondents and conducted a behavioural simulation ( $N=25$ ) to measure how human behaviour overrides technical knowledge.

The findings indicate that technical literacy fails as a defence without habitual security routines. Despite their high proficiency, 100% of the Paradox Group (High Literacy/Low Hygiene) fell victim to a localised lure, mirroring the 100% failure rate of the Vulnerable Group (Low Literacy/Low Hygiene). Conversely, the Cautious Group (Low Literacy/High Hygiene) achieved an 80% resilience rate. By analysing triggers of Authority and Urgency through Protection Motivation Theory (PMT), the study identifies Cyber Hygiene as the essential moderator of digital safety. The research concludes that a single secure habit provides more protection than technical knowledge alone, calling for a shift in UAE policy toward the behavioural habitualization of security.

***Keywords—Cyber Hygiene, Digital Literacy, Social Engineering, Phishing Susceptibility, Protection Motivation Theory, Knowledge-Behaviour Gap, UAE Cybersecurity***

## **CHAPTER I.**

### **THE PROBLEM & ITS BACKGROUND**

This chapter presents the problem and its hypothesis. It also includes the definition of terms involved in the study.

#### **1.1 Introduction**

In the United Arab Emirates, digital transformation is pervasive, with smartphone penetration exceeding 90% and youth navigating a highly connected government ecosystem (TDRA, 2021). However, this technical fluency has created a dangerous irony: the Paradox of Digital Literacy. While "Digital Natives" possess advanced operational skills, the CPX (2024) reports a surge in social engineering attacks, proving that technical proficiency does not equate to security.

Current education assumes that literacy leads to safety, yet emerging research identifies a critical "Knowledge-Behaviour Gap". High literacy often breeds overconfidence, leading users to neglect

essential Cybersecurity practices, such as Multi-Factor Authentication, which ironically makes them more vulnerable to targeted phishing lures (Albladi & Weir, 2020).

This study investigates this disconnect by analysing Cyber Hygiene as a moderating variable. It explores why digital competence fails to translate into secure behaviour. Understanding this relationship is vital for the UAE's National Cybersecurity Strategy, shifting the focus from technical solutions to the "Human Factor", the most vulnerable link in the security chain.

#### **1.2 Statement of the Problem**

Despite the UAE's leading digital infrastructure and the technical fluency of its "Digital Natives," social engineering remains a critical threat. Traditional strategies assume that increasing Digital Literacy (*DL*) reduces vulnerability; however, emerging trends suggest a "Paradox of Digital Literacy," where technical expertise fosters an overconfidence bias that leads users to neglect essential Cyber Hygiene (*CH*).

The core problem is that technical knowledge alone does not prevent susceptibility to psychological manipulation.

There is a lack of empirical evidence in the UAE context regarding the Knowledge-Behaviour Gap, specifically, how habitual hygiene moderates the relationship between knowledge and actual behaviour (Badreddine et al., 2025). This study addresses this gap by investigating why high literacy fails to ensure safety when disconnected from consistent security habits.

### 1.3 Research Objectives

The primary goal of this study is to investigate the "Knowledge-Behaviour Gap" among young adults in the UAE. The specific objectives are:

- ❖ To evaluate current levels of digital literacy and habitual cyber hygiene practices among young adults (ages 18–30) in the UAE.
- ❖ To determine if a correlation exists between high digital literacy and security overconfidence.
- ❖ To measure actual susceptibility to social engineering through localised phishing simulations.
- ❖ To analyse the moderating role of cyber hygiene in the relationship between digital literacy and phishing susceptibility.

### 1.4 Research Questions

This research aims to assess the relationship between digital literacy, cyber hygiene practices, and susceptibility to social engineering attacks among young adults in the United Arab Emirates.

Specifically, this study attempts to answer the following questions:

*RQ1: What is the profile of the participants in terms of:*

*RQ 1.1 Age; and*

*RQ 1.2 Gender;*

*RQ2: What are the self-reported levels of digital literacy and cyber hygiene among young adults in the UAE?*

*RQ3: Is there a negative relationship between high digital literacy and cyber hygiene practices (indicating an overconfidence paradox)?*

*RQ4: How does the frequency of cyber hygiene practices influence an individual's likelihood of falling victim to localised phishing attacks?*

*RQ5: To what degree does cyber hygiene act as a moderating variable between digital literacy and social engineering susceptibility?*

## 1.5 Hypothesis

$H_1$ : There is a relationship between digital literacy and cyber hygiene practices, where higher literacy does not necessarily result in better hygiene (The Paradox).

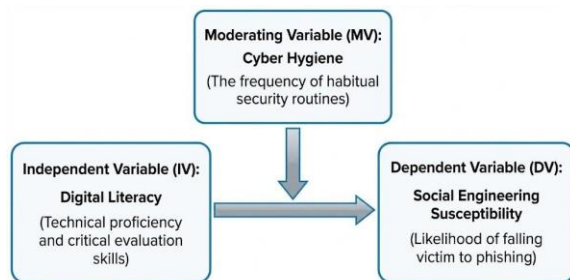
$H_2$ : There is a negative relationship between the quality of cyber hygiene and susceptibility to social engineering attacks.

$H_3$ : Cyber Hygiene ( $CH$ ) moderates the relationship between Digital Literacy ( $DL$ ) and Phishing Susceptibility ( $S$ ), such that high literacy only reduces susceptibility when accompanied by high hygiene scores.

## 1.6 Frameworks

### 1.6.1 Conceptual Framework

*Fig. 1: Conceptual Moderation Model of Phishing Susceptibility ( $DL \times CH \rightarrow S$ )*



As illustrated in Figure 1, the conceptual framework of this study is grounded in the "Paradox of Digital Literacy," proposing that technical proficiency does not have a direct, guaranteed relationship with security. Instead, the relationship is moderated by Cyber Hygiene ( $CH$ ).

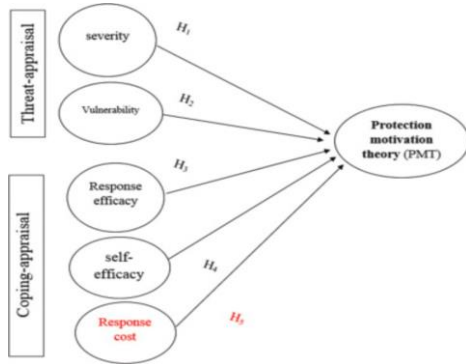
The framework is structured as follows:

- Independent Variable (IV): Digital Literacy (Technical proficiency and critical evaluation skills).
- Moderating Variable (MV): Cyber Hygiene (The frequency of habitual security routines).
- Dependent Variable (DV): Social Engineering Susceptibility (Likelihood of falling victim to phishing).

The model posits that Digital Literacy's effectiveness is contingent upon Cyber Hygiene. If hygiene is low, high literacy may actually increase susceptibility due to Overconfidence. Therefore, literacy serves as a protective shield *only* when paired with high hygiene. This framework tests the "Knowledge-Behaviour Gap" by analysing how the interaction between knowledge and habit dictates final security outcomes.

### 1.6.2 Theoretical Framework

Fig. 2: Theoretical Framework of Protection Motivation Theory (Hedayati et al., 2023)



This study is theoretically grounded in Protection Motivation Theory (PMT) (Rogers, 1975). As illustrated in Figure 2, PMT explains why individuals engage in protective behaviours. In this context, it suggests that a user's motivation to practice *CH* is determined by two cognitive processes:

- Threat Appraisal: An evaluation of the Perceived Severity of a cyberattack and the user's Perceived Vulnerability (Krok et al., 2023).
- Coping Appraisal: An evaluation of Response Efficacy (belief that hygiene works) versus Response Cost (the perceived effort or "friction" required) (Lahiri et al., 2021).

The "Paradox of Digital Literacy" arises when high *DL* leads to a low Threat

Appraisal (via Optimism Bias) and a high Response Cost, effectively neutralising the user's knowledge and increasing their susceptibility (McCombs School of Business, 2013). In this model, *CH* acts as the moderator that determines whether the user's protection motivation translates into actual resilience.

### 1.7 Significance of the Study

This research supports the UAE's "We the UAE 2031" vision for a secure digital society, offering significant value to key stakeholders:

- ❖ For Educators: The findings provide empirical evidence to shift curriculum focus from technical awareness to behavioural habit-formation, addressing the critical "Overconfidence Bias" in technically proficient students.
- ❖ For Policymakers (TDRA/Cybersecurity Council): The data enables the refinement of national campaigns by targeting the "Human Factor." Understanding specific psychological triggers allows for interventions that prioritise active hygiene over passive knowledge.

- ❖ For the Cybersecurity Industry: Contributing to Human-Centric Security, the study offers insights for "security-by-design" interfaces. This encourages developers to reduce "Security Fatigue" and automate hygiene practices for digital natives.
- ❖ For Young Adults: The research serves as a tool for self-reflection, proving that technical fluency does not grant immunity. It encourages a vital shift from reactive overconfidence to proactive behavioural resilience.

### 1.7 Scope and Delimitations

This study is limited to the following scope and aspects:

- ❖ Setting: The study takes place within the digital environment of young adults in the United Arab Emirates, specifically targeting those utilising government and academic digital portals.
- ❖ Participants: The study is limited to a sampled group of approximately 100-150 survey participants and 25 experimental participants, who are young adults (ages 18–30), including university students and graduates.

- ❖ Subject Content: The scope is limited solely to the variables of digital literacy, cyber hygiene, and social engineering susceptibility. It does not account for technical network vulnerabilities or hardware-based security flaws.
- ❖ Academic Year: The study covers the academic year 2025–2026.

### 1.8 Definition of Terms

The following terms are defined conceptually and operationally to provide a clear understanding of the study:

*Digital Literacy (DL).* Conceptually, the ability to find, evaluate, and communicate information through digital platforms. Operationally, it is measured using the Hwang et al. (2023) scale.

*Cyber Hygiene (CH).* The routine, habitual practices users perform to maintain security (e.g., MFA, updates). Operationally measured via the Jansen and van Schaik (2020) scale.

*Social Engineering Susceptibility (S).* A binary behavioural metric (1 for clicked/consented, 0 for resisted) measuring a user's response to the "Ellen" phishing lure.

*Knowledge-Behaviour Gap.* The phenomenon where high scores in Digital Literacy do not correlate with a high frequency of Cyber Hygiene practices.

*Response Cost.* The perceived level of effort, time, or "friction" required to implement security measures (e.g., the annoyance of entering an SMS code for MFA).

*Response Inefficacy.* The belief that individual security actions (like changing a password) are useless against sophisticated hackers.

*Cognitive Bypass.* A state where psychological triggers (Urgency or Authority) cause a user to stop thinking critically and react instinctively to a phishing lure.

## CHAPTER II.

### LITERATURE REVIEW

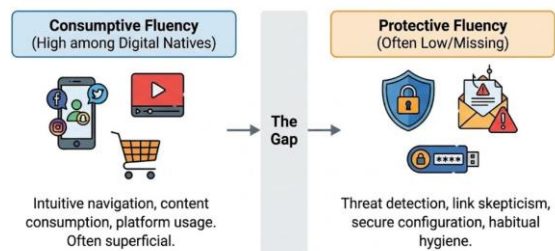
This chapter reviews the theoretical foundations and existing literature surrounding the Paradox of Digital Literacy.

#### 2.1 The “Digital Native” Fallacy and the Paradox of Digital Literacy

The term "Digital Native" (Creighton, 2018; Prensky, 2001) suggests inherent technical

fluency, a label often applied to UAE youth. However, empirical evidence reveals this proficiency is largely "consumptive", focused on application usage rather than "protective" threat detection (Hwang et al., 2023). As illustrated in Figure 3, this discrepancy creates a dangerous risk profile described as a "false sense of security" (Albladi & Weir, 2020).

Fig. 3: The “Digital Native” Skill Gap: Consumptive vs. Protective Fluency



Rooted in Overconfidence Theory, digital natives frequently underestimate social engineering threats, assuming their navigational ease equates to scam detection intuition (EBSCO, 2025). Furthermore, the Paradox of Digital Literacy implies that seamless User Experience (UX) in platforms like UAE PASS actually diminishes security understanding. As technical barriers vanish, users become detached from the mechanisms protecting their data, leading to a decline in active cyber hygiene. Consequently, Jansen and van Schaik (2020) argue that modern literacy must be redefined

not as the ability to use technology, but the ability to secure it.

## **2.2. The Knowledge-Behaviour Gap and Protection Motivation Theory**

The transition from digital literacy (knowing) to cyber hygiene (doing) is non-linear, a disconnect best explained by the Knowledge-Behaviour Gap and Protection Motivation Theory (Rogers, 1975). PMT posits that protective intention relies on two cognitive processes: Threat Appraisal and Coping Appraisal (Marikyan and Papagiannidis, 2023).

In the UAE student population, Threat Appraisal is often suppressed by Optimism Bias (Alnifie and Kim, 2023), where young adults view themselves as unlikely targets compared to corporations or older generations. Without perceived vulnerability, motivation to implement practices like MFA remains stagnant despite technical knowledge (Schaik, 2017).

Simultaneously, Coping Appraisal is undermined by Response Cost, the perceived "friction" of security measures (Folkman et al., 1986). For digital natives prioritising convenience, the annoyance of complex passwords often outweighs the benefits (Zimmermann and Renaud, 2019).

Furthermore, Response ineffectiveness leads students to believe individual actions cannot stop sophisticated attacks (Baraković and Baraković Husić, 2022; Mou et al., 2022). Consequently, literacy alone is a poor predictor of safety; rather, it is the user's psychological evaluation of the "cost" of protection that dictates their vulnerability.

## **2.3 Defining Cyber Hygiene: from Technical Task to Behavioural Routine**

Cyber hygiene refers to habitual individual practices (like MFA) maintained to prevent systemic infection, analogous to personal hygiene. However, a critical gap exists between knowledge and compliance due to "Security Fatigue". In the UAE, the friction of security conflicts with convenience, causing even literate users to ignore alerts and prioritise time-saving defaults (Jibin, 2025). This weariness effectively neutralises technical literacy, increasing vulnerability despite expertise.

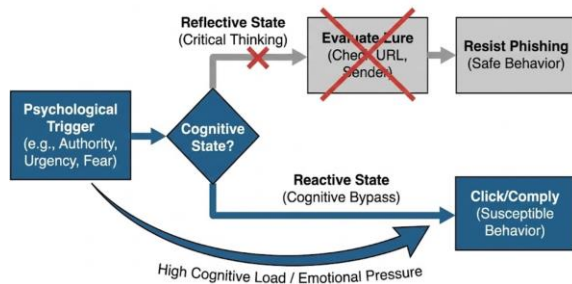
## **2.4 Social Engineering Susceptibility: the Exploitation of Human Heuristics**

Social engineering is defined as the manipulation of individuals into divulging confidential information (Albladi & Weir, 2020), utilising psychological levers like



Authority and Urgency to bypass digital literacy. As illustrated in Figure 4, attackers in the UAE often localise these threats, mimicking trusted entities like UAE PASS, to trigger a Cognitive Bypass. This psychological pressure forces users from a reflective to a reactive state, temporarily neutralising their technical knowledge (Saeed et al., 2023).

*Fig. 4: The Cognitive Bypass Model of Phishing Susceptibility*



Susceptibility is further exacerbated by Cognitive Load. In a hyper-connected student population, multitasking strains mental resources, degrading the ability to critically evaluate sophisticated lures (Global Council, 2025). This confirms the necessity of treating Cyber Hygiene as a moderator. In this model, Digital Literacy acts as a shield *only* when paired with high hygiene habits. Conversely, low hygiene neutralises technical proficiency, leaving even expert users vulnerable to emotional manipulation regardless of their skills.

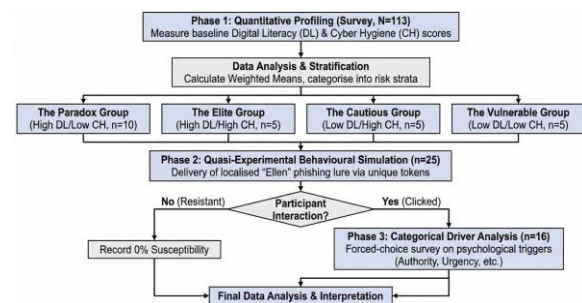
## CHAPTER III: METHODOLOGY

This chapter details the systematic approach used. It outlines the research design, participant stratification, the development of the localised phishing simulation, and the ethical safeguards employed

### 3.1 Research Design

This study adopts a Multi-Phase Quantitative Research Design to investigate the relationship between technical knowledge and security behaviour. By utilising a purely quantitative framework, the study bridges the gap between self-reported psychometric data and observed behavioural outcomes through a structured, three-phase approach (Banawa, 2025) (see Figure 5):

*Fig. 5: Systematic Multi-Phase Research and Stratification Workflow*



1. *Quantitative Profiling (Phase 1):* A descriptive cross-sectional survey

used to establish baseline levels of Digital Literacy (*DL*) and Cyber Hygiene (*CH*) among the participants.

2. *Quasi-Experimental Simulation (Phase 2)*: A controlled behavioural experiment designed to measure real-world phishing susceptibility (*S*) against a localised stimulus.
3. *Categorical Driver Analysis (Phase 3)*: A forced-choice quantitative follow-up to identify and categorise the specific psychological heuristics, such as Authority or Urgency, that triggered participant engagement.

### **3.1.1 Quasi-Experimental Non-Equivalent Group Design**

Within Phase 2, a Quasi-Experimental Non-Equivalent Groups Design was utilised. Unlike a true laboratory experiment, which relies on random assignment, this study assigns participants to specific "risk strata" based on their pre-existing *DL* and *CH* scores captured in Phase 1.

This design is essential because Digital Literacy and Cyber Hygiene are inherent behavioural traits that cannot be manipulated or artificially induced by the researcher. By comparing these non-equivalent groups, specifically The

Paradox, The Elite, The Cautious, and The Vulnerable, against the same controlled stimulus, the study provides a quantitative observation of how distinct combinations of knowledge and habit influence final security outcomes (Krishnan, 2019).

## **3.2 Participants and Sampling**

### **3.2.1 Stage 1: the Initial Survey Pool**

The study utilised Snowball Sampling via digital networks. The initial research link was distributed through institutional WhatsApp groups in the UAE, with participants encouraged to share the link with their peers. This resulted in a pool of 113 respondents. The survey captured consent, demographics (age, gender, email), and baseline psychometric scores.

### **3.2.2 Stage 2: Disproportionate Stratified Sampling**

From the 113 respondents, Disproportionate Stratified Sampling was used to select 25 participants for the behavioural simulation. This method ensured the "Paradox Group" was oversampled to provide sufficient data for the study's primary focus.

Table 3.1: Stratified Experimental Matrix (N=25)

| Group Category       | DL Criteria (Mean)  | CH Criteria (Mean)   | Sample Size (N) |
|----------------------|---------------------|----------------------|-----------------|
| The Paradox Group    | High ( $\geq 4.0$ ) | Lower ( $\leq 2.9$ ) | 10              |
| The Elite Group      | High ( $\geq 4.0$ ) | High ( $\geq 4.0$ )  | 5               |
| The Cautious Group   | Low ( $\leq 2.9$ )  | High ( $\geq 4.0$ )  | 5               |
| The Vulnerable Group | Low ( $\leq 2.9$ )  | Low ( $\leq 2.9$ )   | 5               |

### 3.3 Instrumentation

The research instrument was a multi-stage survey questionnaire comprising validated psychometric scales and forced-choice behavioural questions.

#### 3.3.1 Quantification and Likert Scale Legend

All psychometric responses from Phase 1 were assigned a numerical weight to calculate the Weighted Mean (WM) for each participant.

To simplify the analysis of participant responses, the five-point Likert scale was grouped into four primary interpretive categories. This categorisation helps in clearly identifying "High" versus "Low" proficiency and habit levels.

Table 3.2: Likert Scale Legend and Categorical Table

| Scale | Range of Values | Categorical Interpretation | Verbal Description (DL) | Verbal Description (CH) |
|-------|-----------------|----------------------------|-------------------------|-------------------------|
| 5     | 4.01 – 5.00     | High                       | Technical Proficiency   | Frequent / Always       |
| 4     | 3.01 – 4.00     | Average                    | Moderate Literacy       | Occasional / Sometimes  |
| 3     | 2.01 – 3.00     | Low                        | Limited Literacy        | Rare / Infrequent       |
| 2 – 1 | 1.00 – 2.00     | Very Low                   | Deficient Literacy      | Never / Non-existent    |

#### 3.3.2 The Psychometric Scales

- ❖ Digital Literacy (DL): This scale was adapted from the Digital Literacy Assessment Framework developed by Hwang, Shin, and Kim (2023). It measures Technical Proficiency,

Information Evaluation, Privacy Awareness, Risk Detection, and Critical Thinking within digital ecosystems.

- ❖ Cyber Hygiene (*CH*): This scale was adapted from the Cyber Hygiene Inventory (*CH*) developed by Jansen and van Schaik (2019). It assesses the frequency of habitual protective behaviours, including password complexity, multi-factor authentication (MFA) adoption, software update frequency, and link scepticism.

### 3.4 The "Ellen" Phishing Protocol

The behavioural phase utilised a localised social engineering lure to measure real-world susceptibility.

1. The Stimulus: A localised email from "Ellen | Admin Department" (infodesk.it.ae@gmail.com) was sent to the 25 participants.
2. Tokenisation: Each participant was assigned a unique reference token (e.g., J01 to J25) appended to their specific link (?token=JXX).
3. The Landing Page: Clicking the link led to a Research Debriefing Page on GitHub. The page immediately

informed participants they were part of a simulation.

4. Forced-Choice Data Capture: Participants were given a "YES/NO" option to include their data. Those who proceeded selected their reason for clicking from a predetermined list (Authority, Urgency, Multi-tasking, Trust in UAE Notifications, or Curiosity).

### 3.5 Ethical Considerations

The study adhered to the Bath Spa University Research Ethics Policy:

- ❖ Active Deception: To ensure the authenticity of the behavioural response and avoid the "Hawthorne Effect" (where participants change their behaviour because they know they are being watched), active deception was utilised. This involved the creation of a fabricated administrative persona ("Ellen") and a localised pretext.
- ❖ Informed Consent: Obtained at Phase 1 and re-verified at the point of the "phish" interaction.
- ❖ Right to Withdraw: Participants could exit by closing the tab or entering "Do Not Consent" in the final form. In such cases, their

interaction was recorded as a '0' to protect participant autonomy.

- ❖ Anonymity: Data was linked only to pseudonymous J-tokens.

### 3.6 Methodological Limitations

The primary flaw of the Non-Equivalent Groups Design is Selection Bias (Masterclass, 2020). Since the researcher did not randomly assign participants, unobserved variables (e.g., stress) could act as confounding factors. Furthermore, the reliance on self-reported data in Phase 1 introduces Social Desirability Bias, where participants may over-report hygiene habits to appear more security-conscious.

### 3.7 Data Analysis Plan

Data collected from the 113 survey respondents and 25 experimental participants were analysed using descriptive statistics. The Weighted Mean (WM) was calculated for each psychometric variable to categorise participants into risk strata. Susceptibility rates (*S*) were calculated as a percentage of successful clicks per group. Finally, categorical data from Phase 3 were analysed using frequency distribution to determine the primary psychological drivers of engagement.

## CHAPTER IV: PRESENTATION, ANALYSIS, AND INTERPRETATION OF DATA

This chapter presents the empirical results of the study, organised according to the research questions.

### 4.1 Demographic Profile of Participants

The study captured data from a total of 113 respondents across the United Arab Emirates. Table 4.1 illustrates the distribution of the sample in terms of gender and age, confirming the study's focus on the "Digital Native" demographic.

*Table 4.1: Frequency and Percentage Distribution of Participants (N=113)*

| Variable | Category          | Frequency (f) | Percentage (%) |
|----------|-------------------|---------------|----------------|
| Gender   | Female            | 56            | 49.6%          |
|          | Male              | 54            | 47.8%          |
|          | Prefer not to say | 3             | 2.6%           |
| Age      | 18 – 21           | 86            | 76.1%          |

|              |         |     |       |
|--------------|---------|-----|-------|
|              | 22 – 25 | 20  | 17.7% |
|              | 26 – 30 | 7   | 6.2%  |
| <b>Total</b> |         | 113 | 100%  |

Interpretation: The demographic data reveals a balanced gender distribution, minimising sex-based bias in the results. The age distribution is heavily concentrated in the 18–21 bracket (76.1%), ensuring that the findings reflect the behaviours of university-aged digital natives who are most active within the UAE’s digital ecosystem.

#### 4.2 Baseline Levels of Literacy and Hygiene

To address RQ2, the study measured the self-reported levels of Digital Literacy (*DL*) and Cyber Hygiene (*CH*).

*Table 4.2: Summary of Grand Weighted Means*

| Variable                            | Grand Weighted Mean | Verbal Interpretation |
|-------------------------------------|---------------------|-----------------------|
| <b>Digital Literacy (<i>DL</i>)</b> | 3.87                | High                  |

|                                  |      |              |
|----------------------------------|------|--------------|
| <b>Cyber Hygiene (<i>CH</i>)</b> | 3.73 | High / Often |
|----------------------------------|------|--------------|

Interpretation: Both variables scored within the "High" range. However, the slightly higher score for *DL* (3.87) compared to *CH* (3.73) suggests that while participants feel technically competent, their consistent application of security routines lags slightly behind their perceived knowledge. This sets the stage for the "Knowledge-Behaviour Gap" observed in the experimental phase.

#### 4.3 Results of the "Ellen" Phishing Simulation

To address RQ3 and RQ4, a stratified sample of 25 participants was subjected to the localised phishing lure.

*Table 4.3: Comparative Susceptibility Rates by Group (N=25)*

| Group Category           | Sample Size ( <i>N</i> ) | Number of Clicks | Susceptibility Rate (%) |
|--------------------------|--------------------------|------------------|-------------------------|
| <b>The Paradox Group</b> | 10                       | 10               | 100%                    |

|                             |   |   |      |
|-----------------------------|---|---|------|
| <b>The Vulnerable Group</b> | 5 | 5 | 100% |
| <b>The Cautious Group</b>   | 5 | 1 | 20%  |
| <b>The Elite Group</b>      | 5 | 0 | 0%   |

#### 4.3.1 Comparative Analysis of Resilience and Susceptibility

The results provide striking evidence for the Paradox of Digital Literacy:

- The Paradox Failure: Despite the highest self-reported proficiency, 100% of the Paradox Group fell victim to the lure, confirming that technical skill is a poor shield without a behavioural routine.
- The Elite Resilience: Conversely, the Elite Group achieved 0% susceptibility, suggesting literacy only works when paired with high-frequency hygiene.
- The Cautious Success: Most notably, the Cautious Group (Low Literacy/High Hygiene) demonstrated an 80% resilience rate, proving that habitual hygiene is a

superior predictor of safety than technical knowledge alone.

#### 4.4 Analysis of Behavioural Drivers

For the 16 participants who engaged with the lure, the study utilised a forced-choice instrument to identify the primary psychological trigger.

*Table 4.4: Frequency of Categorical Drivers for Engagement (N=16)*

| Driver           | Frequency (f) | Percentage (%) |
|------------------|---------------|----------------|
| <b>Authority</b> | 7             | 43.75%         |
| <b>Urgency</b>   | 6             | 37.50%         |
| <b>Curiosity</b> | 3             | 18.75%         |
| <b>Total</b>     | 16            | 100%           |

Interpretation: Authority (43.75%) emerged as the primary heuristic, confirming high susceptibility to lures mimicking institutional power. Urgency (37.5%) followed, indicating that pressure successfully triggers a "Cognitive Bypass." Notably, environmental factors like multi-tasking received zero votes, suggesting that the lure's psychological

content was significantly more influential than situational context or platform trust.

#### 4.5 Synthesis: The Knowledge-Behaviour Gap

The data confirms a significant Knowledge-Behaviour Gap. The fact that the Cautious Group outperformed the Paradox Group despite having "lower" literacy scores suggests that cybersecurity in the UAE must shift its focus.

The 100% failure rate of the Paradox Group versus the 0% failure rate of the Elite group identifies Cyber Hygiene (*CH*) as the critical moderating variable. Without *CH*, digital literacy is merely passive knowledge that fails under the psychological pressure of authority and urgency.

### CHAPTER V: SUMMARY, CONCLUSION, AND RECOMMENDATIONS

This final chapter offers a definitive conclusion on the moderating role of Cyber Hygiene and presents actionable recommendations for improving the cybersecurity posture of the "Digital Native" demographic in the United Arab Emirates.

#### 5.1 Summary of Findings

The study utilised a dual-stage quantitative framework ( $N=113$ ) to investigate the gap between technical knowledge and behavioural resilience.

- ❖ **Psychometric Profiling ( $N=113$ ):** Participants reported high Digital Literacy (WM 3.87) and positive Cyber Hygiene (WM 3.73), establishing the baseline for the observed gap.
- ❖ **The Behavioural Paradox ( $N=25$ ):** The experimental phase revealed a critical disconnect: the "Paradox Group" (High Literacy/Low Hygiene) suffered a **100% susceptibility rate** to the localised lure.
- ❖ **The Primacy of Habit:** Conversely, the Elite (**0%**) and Cautious (**20%**) groups demonstrated that consistent hygiene provides superior defence compared to technical literacy alone.
- ❖ **The Power of Heuristics:** **Authority (43.75%)** and **Urgency (37.50%)** emerged as the dominant triggers, successfully bypassing technical knowledge to induce immediate engagement



## 5.2 Conclusion

The central conclusion is that Cyber Hygiene (*CH*) acts as a critical moderator between Digital Literacy and Susceptibility. Empirical evidence debunks the "Digital Native" fallacy, proving that technical literacy does not equate to security.

Applying Protection Motivation Theory, the Paradox Group's failure reflects a breakdown in Threat Appraisal. High literacy fosters "Optimism Bias," reducing perceived vulnerability and causing users to neglect protocols under pressure. Conversely, the Cautious Group's resilience validates Coping Appraisal, specifically habitual routine, as the superior safety predictor. In the UAE, technical knowledge is merely a passive asset; without the "active shield" of hygiene, even experts remain vulnerable to emotional manipulation.

## 5.3 Recommendations

### 5.3.1 For UAE Educational Institutions

- ❖ From Awareness to Habitualization: Universities must transition from passive lectures to active habit formation. The Paradox Group's 100% failure rate confirms that knowledge is insufficient. Institutions should implement

mandatory tools, such as password managers or hardware-based MFA, to reduce the Response Cost of staying secure.

- ❖ Active Simulation Programs: Leveraging the "Ellen" lure's success, universities should conduct regular, unannounced phishing simulations. These must serve as non-disciplinary "learning moments," providing immediate categorical feedback to help students recognise their susceptibility to Authority and Urgency triggers.

### 5.3.2 For UAE Cybersecurity Policy (TDRA)

- ❖ Public Persona Protection: Since Authority was the top driver, the TDRA should launch a targeted campaign titled "*The Admin Fallacy*." This campaign should educate the public on how attackers spoof institutional identities and emphasise that no legitimate UAE entity will use urgency to bypass security protocols.
- ❖ Human-Centric Design: Security developers in the region should focus on "Zero-Friction Hygiene." By making secure behaviours the default

setting in apps like UAE PASS, the burden of protection is removed from the user's overconfident technical judgment.

#### **5.4 Limitations and Suggestions for Future Research**

The most significant limitation of this research is the experimental sample size ( $N=25$ ). While the 100% failure rate of the Paradox Group is a compelling trend, it represents a "pilot" observation. Because the assignment was Non-Equivalent, there is a possibility of "Selection Bias" where unobserved factors like the participants' stress levels during the email delivery could have influenced the "Click."

Future Research Should:

1. Introduce Qualitative Inquiry: A follow-up phase of semi-structured interviews is recommended to ask the one "Cautious" participant *why* they clicked, and the "Elite" participants *how* they spotted the fake "Ellen" persona. This would add depth to the current quantitative data.
2. Longitudinal Study: Research should investigate if a "Phishing Failure" leads to a permanent increase in

Cyber Hygiene scores over a 6-month period.

3. Cross-Demographic Analysis: Future studies should compare the "Paradox of Digital Literacy" in young adults against older "Digital Immigrants" to see if lower literacy actually leads to higher caution across all age groups in the UAE.

#### **6. References**

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